SE LAB-5

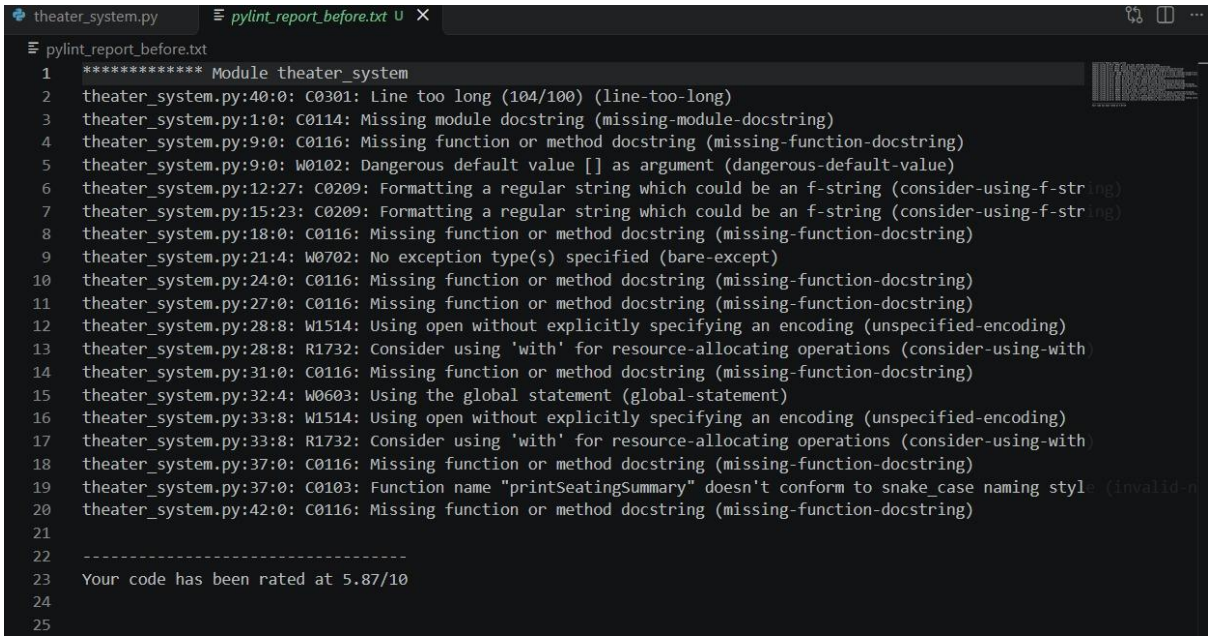
Name: Sanjith R

SRN: PES2UG24CS445

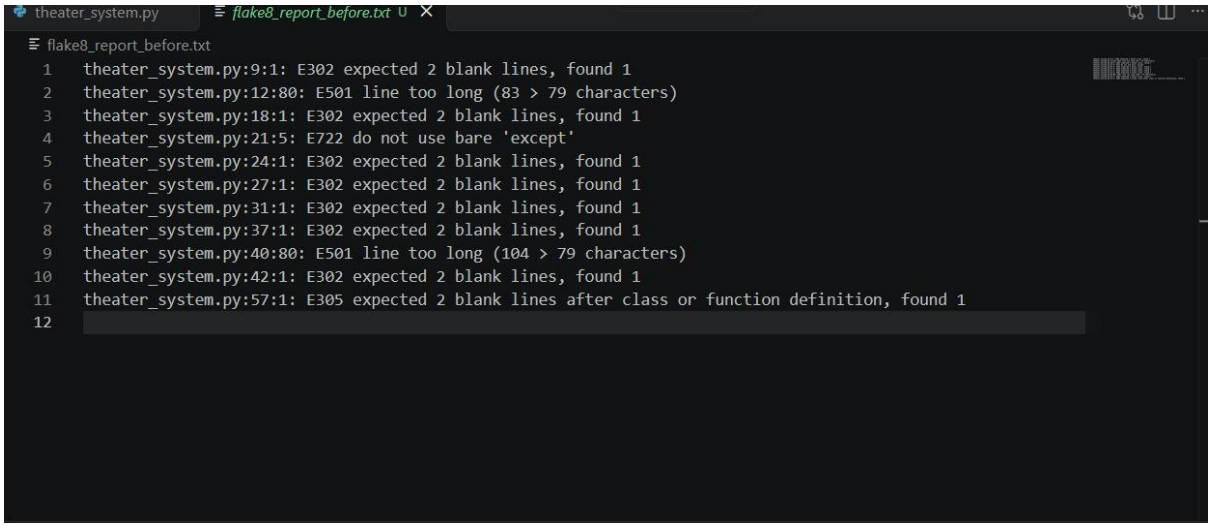
SEC: H

BEFORE CHANGES:

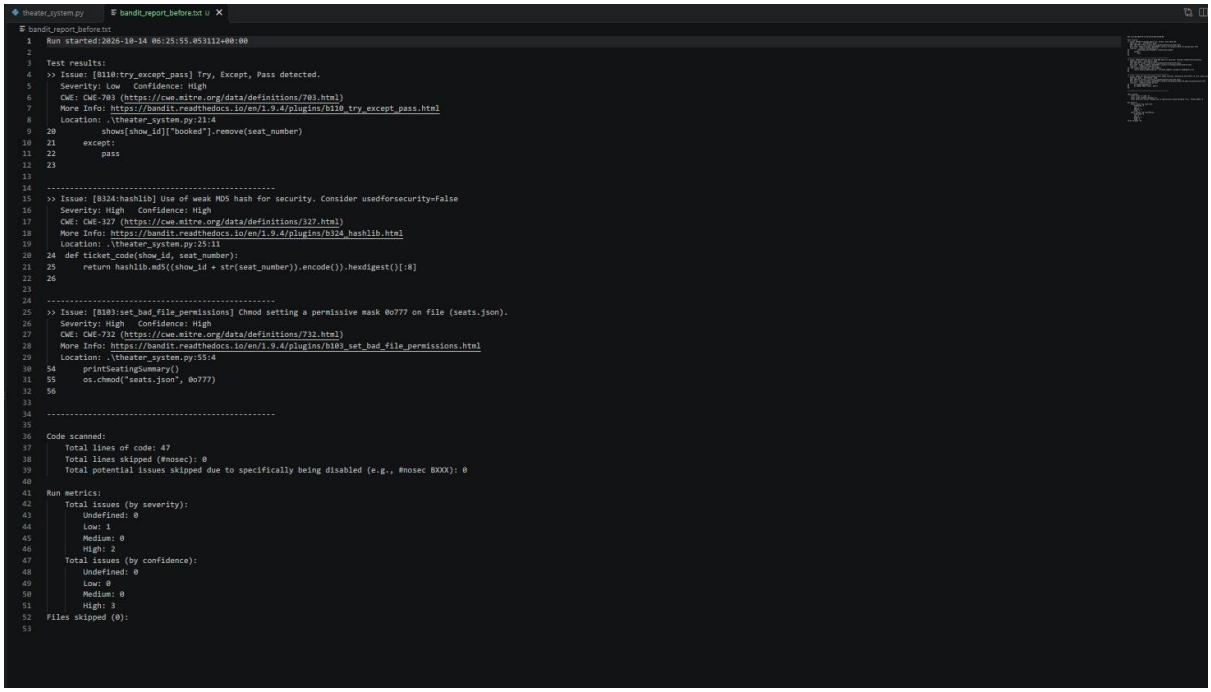
pylint_report_before



flake8_before



bandit_before



```
Run started:2026-10-14 06:25:55.053112+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\theater_system.py:21:4
20          shows[show_id]["booked"].remove(seat_number)
21      except:
22          pass
23

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\theater_system.py:25:11
24  def ticket_code(show_id, seat_number):
25      return hashlib.md5((show_id + str(seat_number)).encode()).hexdigest()[:8]
26

--------------------------------------------------
>> Issue: [B103:set_bad_file_permissions] Chmod setting a permissive mask 0o777 on file (seats.json).
   Severity: High   Confidence: High
   CWE: CWE-732 (https://cwe.mitre.org/data/definitions/732.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b103_set_bad_file_permissions.html
   Location: .\theater_system.py:55:4
54      printSeatingSummary()
55      os.chmod("seats.json", 0o777)
56

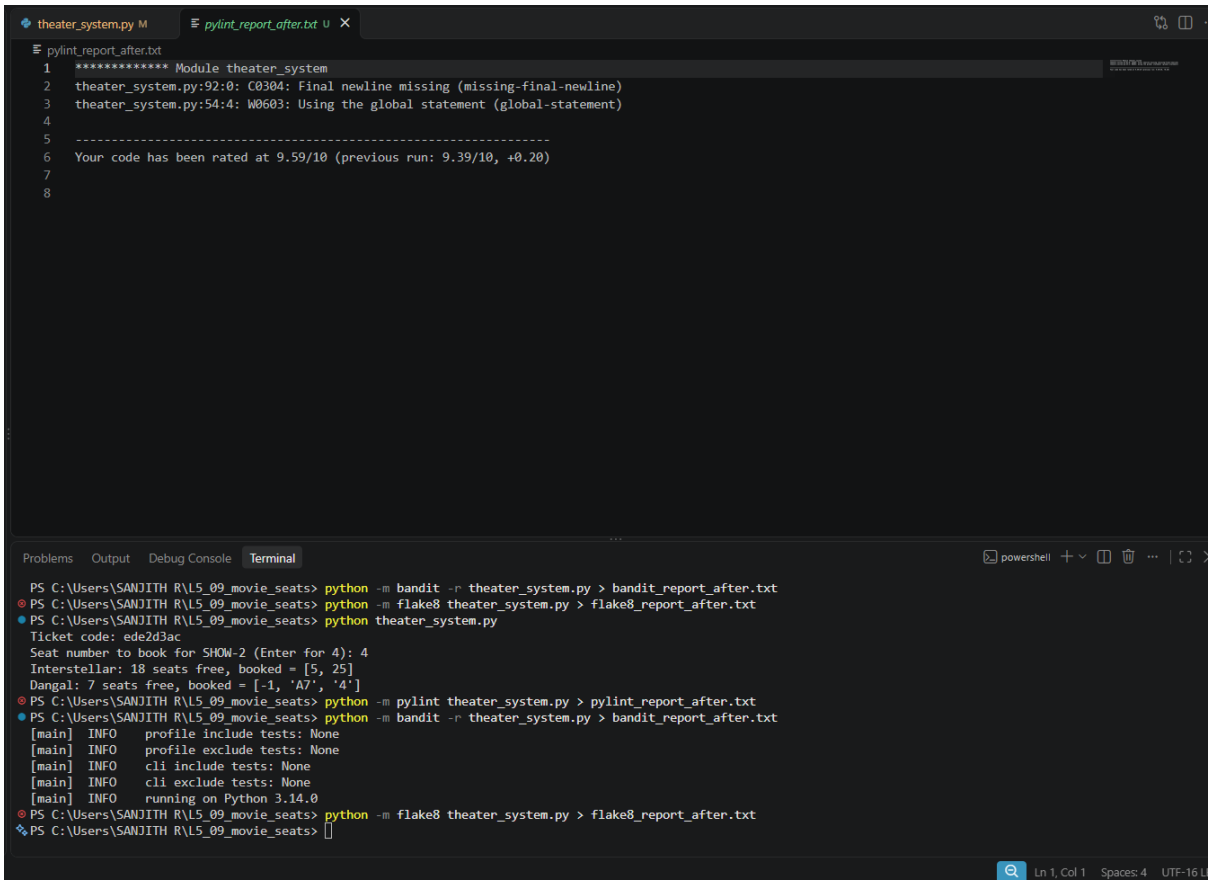
--------------------------------------------------

Code scanned:
    Total lines of code: 47
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 1
        Medium: 0
        High: 2
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 3
Files skipped (0):
```

AFTER CHANGES:

pylint_report_after



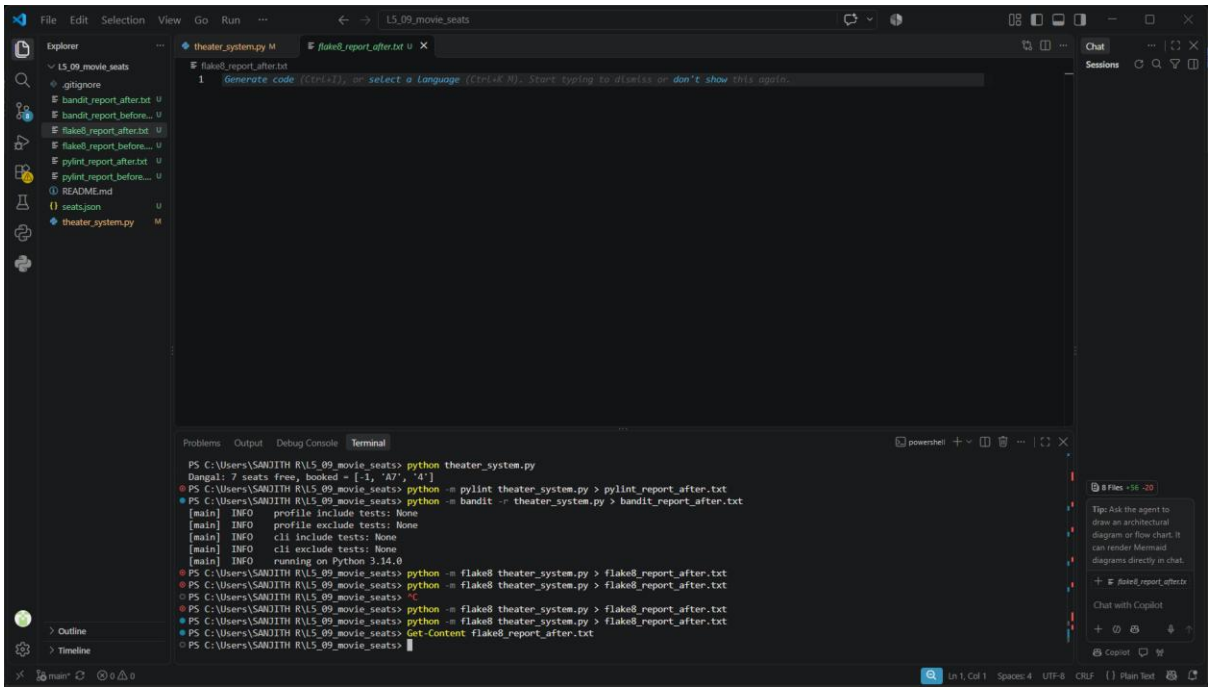
```
theater_system.py M   pylint_report_after.txt U
pylint_report_after.txt
1  ************* Module theater_system
2  theater_system.py:92:0: C0304: Final newline missing (missing-final-newline)
3  theater_system.py:54:4: W0603: Using the global statement (global-statement)
4
5  -----------------------------------------------------------------
6  Your code has been rated at 9.59/10 (previous run: 9.39/10, +0.20)
7
8

Problems  Output  Debug Console  Terminal                    powershell

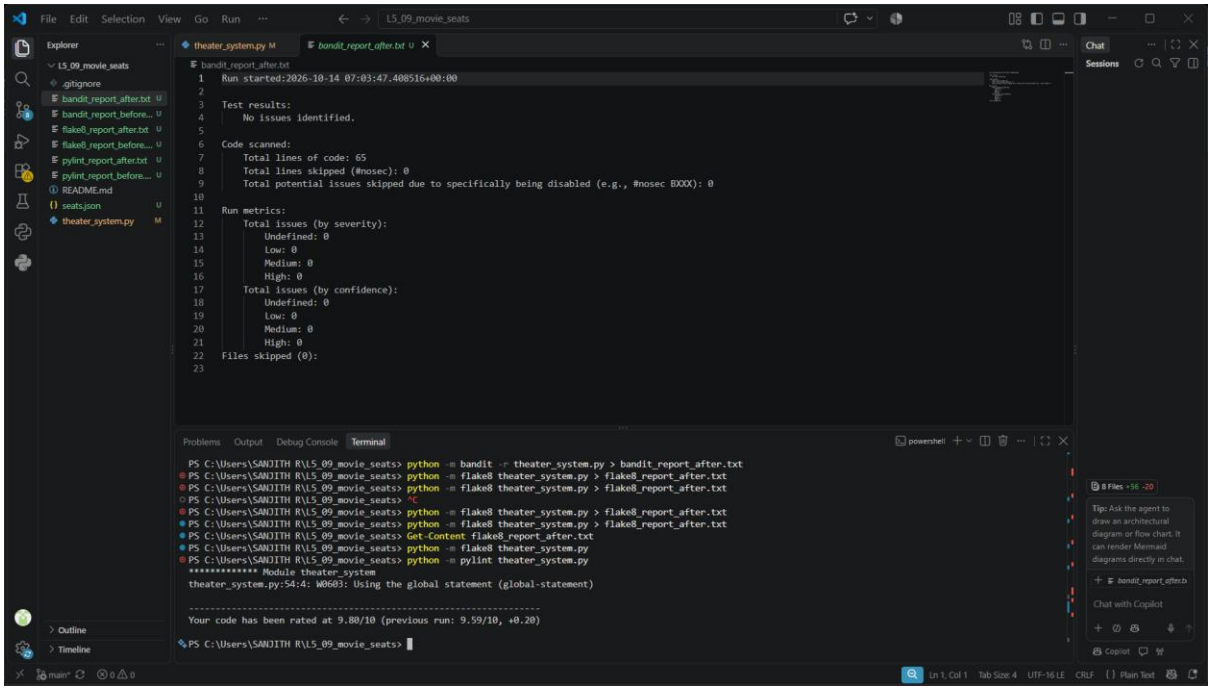
PS C:\Users\SANJITH R\L5_09_movie_seats> python -m bandit -r theater_system.py > bandit_report_after.txt
PS C:\Users\SANJITH R\L5_09_movie_seats> python -m flake8 theater_system.py > flake8_report_after.txt
PS C:\Users\SANJITH R\L5_09_movie_seats> python theater_system.py
Ticket code: ede2d3ac
Seat number to book for SHOW-2 (Enter for 4): 4
Interstellar: 18 seats free, booked = [5, 25]
Dangal: 7 seats free, booked = [-1, 'A7', '4']
PS C:\Users\SANJITH R\L5_09_movie_seats> python -m pylint theater_system.py > pylint_report_after.txt
PS C:\Users\SANJITH R\L5_09_movie_seats> python -m bandit -r theater_system.py > bandit_report_after.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.0
PS C:\Users\SANJITH R\L5_09_movie_seats> python -m flake8 theater_system.py > flake8_report_after.txt
PS C:\Users\SANJITH R\L5_09_movie_seats>

Ln 1, Col 1   Spaces: 4   UTF-16 L
```

Flake8_report_after



Bandfit_report_after



Python_code_after_changes:

"""Manage movie theater seat reservations and ticket codes."""

```
import hashlib
import json
import os

# Global state
shows = {
    "SHOW-1": {"title": "Interstellar", "seats": 20, "booked": []},
    "SHOW-2": {"title": "Dangal", "seats": 10, "booked": []},
}

def reserve_seat(show_id, seat_number, system_logs=None):
    """Reserve a seat and record the operation."""
    if system_logs is None:
        system_logs = []

    show = shows[show_id]

    if seat_number in show["booked"]:
        system_logs.append(
            f"Seat {seat_number} already taken for {show_id}"
        )
        return False

    show["booked"].append(seat_number)
    system_logs.append(f"Reserved seat {seat_number} for {show_id}")
    return True

def release_seat(show_id, seat_number):
```

```
    """Release a booked seat if it exists."""
    try:
        shows[show_id]["booked"].remove(seat_number)
    except (KeyError, ValueError):
        pass

def ticket_code(show_id, seat_number):
    """Generate a ticket code using SHA-256."""
    ticket_data = f"{show_id}{seat_number}".encode()
    return hashlib.sha256(ticket_data).hexdigest()[:8]

def dump_reservations(file="seats.json"):
    """Save reservations to a JSON file."""
    with open(file, "w", encoding="utf-8") as reservation_file:
        json.dump(shows, reservation_file)

def load_reservations(file="seats.json"):
    """Load reservations from a JSON file."""
    global shows

    with open(file, "r", encoding="utf-8") as reservation_file:
        shows = json.load(reservation_file)

def print_seating_summary():
    """Print free and booked seats for each show."""
    for show in shows.values():
        free = show["seats"] - len(show["booked"])
        print(
            f"{show['title']}: {free} seats free, "
```

```
            f"booked = {show['booked']}"
        )

def main():
    """Run a demonstration of the theater reservation system."""
    reserve_seat("SHOW-1", 5)
    reserve_seat("SHOW-1", 5)
    reserve_seat("SHOW-1", 25)
    reserve_seat("SHOW-2", -1)
    reserve_seat("SHOW-2", "A7")
    release_seat("SHOW-2", 3)

    print("Ticket code:", ticket_code("SHOW-1", 5))

    seat = input("Seat number to book for SHOW-2 (Enter for 4): ") or "4"
    reserve_seat("SHOW-2", seat)

    dump_reservations()
    load_reservations()
    print_seating_summary()

    os.chmod("seats.json", 0o600)

if __name__ == "__main__":
    main()
```

Before and After Code Analysis – Movie Seat Reservation System

NO	Tool	Before	After	Status
1	**Pylint**	**Score: 5.87/10**	**Score:9.59/10**	**Verify score is above 9.5**

2	**Flake8**	**Multiple formatting violations**	**No violations**	**Verify**
3	**Bandit**	**3 security findings**	**No issues identified**	**Passed**